



Geometry Honors

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADES 8–10 · FULL YEAR · HONORS

~4¼ hrs per week · 50 min/day, M–F

About this overview. This document is an example of how the course might be taught. Scholars may see some variation in lesson order, examples, and activities from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Geometry Honors invites scholars to study the order and beauty inherent in space itself. Through shapes, proofs, and theorems, students develop the habits of precise reasoning and argumentation that define mathematics. In immersive VR- and AI-enhanced lessons, scholars engage five areas: proving and applying theorems about two-dimensional figures using Euclidean and coordinate geometry; establishing congruence and similarity through Euclidean criteria and rigid transformations; extending geometric measurement to two- and three-dimensional figures; creating and applying equations of circles in the coordinate plane; and developing an understanding of right triangle trigonometry. As an honors course, the work is more abstract and demanding throughout. Prerequisite: Algebra 1.

THE YEAR AT A GLANCE

1 Proof in Two Dimensions (area 1)

Proving and applying relationships and theorems about 2-D figures using Euclidean and coordinate geometry.

3 Geometric Measurement (area 3)

Extending measurement to two- and three-dimensional figures — area, surface area, volume, and cross-sections.

5 Right Triangle Trigonometry (area 5)

Defining and applying sine, cosine, and tangent, including the special right triangles.

2 Congruence and Similarity (area 2)

Establishing congruence and similarity using Euclidean criteria and rigid transformations.

4 Circles in the Coordinate Plane (area 4)

Creating and applying equations of circles, with arcs, sectors, chords, and tangents.

The Honors difference. Greater rigor throughout — proof by contradiction, the Laws of Sines and Cosines, additional compass-and-straightedge constructions, and formal justification of the congruence and similarity criteria. This prepares scholars directly for Algebra 2 Honors and the honors mathematics sequence.

A TYPICAL WEEK

- A new lesson most days: teacher-guided instruction with worked examples and notes (about 30–35 min).
- Practice problems with every lesson, worked out on paper, to make each skill automatic (about 15–20 min).
- VR and AI tools make spatial relationships and three-dimensional transformations visible; quizzes and tests close each unit.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Daily lesson + practice fills most of the 50 minutes.
- Module review and assessment days replace new lessons as each module closes.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- A dedicated math notebook, pencil, and paper — every problem gets worked by hand.
- A scientific or graphing calculator, plus compass and straightedge for constructions.
- Microsoft 365 and OneDrive (provided to every scholar) for any typed work.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.